Maintenance Revenue

Year 2019 compared with year 2018

Maintenance revenue increased $2.2 million in 2019 compared to 2018 primarily due to (1) an increase in the first-year maintenance revenue; (2) our annual renewal rate of customers subscribing to maintenance, which was greater than 90%; and (3) increases in the maintenance renewal prices. Maintenance revenue for the Americas, EMEA and APAC segments increased $1.4 million, $0.4 million and $0.4 million, respectively, compared to 2018.

Year 2018 compared with year 2017

Maintenance revenue increased $4.0 million in 2018 compared to 2017 primarily due to (1) an increase in the first-year maintenance revenue; (2) our annual renewal rate of customers subscribing to maintenance, which was greater than 90%; and (3) increases in the maintenance renewal prices. Maintenance revenue for the Americas, EMEA and APAC segments increased $1.1 million, $2.2 million and $0.7 million, respectively, compared to 2017.

Services Revenue

Year 2019 compared with year 2018

Services revenue increased $30.8 million, or 9%, in 2019 compared to 2018. The Americas, EMEA and APAC segments increased $17.8 million, $10.3 million and $2.7 million, respectively, compared to 2018.

Year 2018 compared with year 2017

Services revenue increased $3.2 million in 2018 compared to 2017 primarily due to improving demand in the Americas and solid growth in EMEA. Services revenue for the Americas and EMEA segment increased $1.0 million and $6.9 million, respectively, and services revenue for the APAC segment decreased $4.7 million, compared to 2017.

Hardware

Hardware sales, net decreased $1.5 million, or -10% in 2019 compared to 2018. We adopted the new ASC 606 standard as of January 1, 2018 and elected to use the modified retrospective method. Historical hardware sales prior to the adoption of ASC 606 were recorded on a gross basis, as we were the principal in the transaction in accordance with the previous standard, ASC 605-45. Under the new standard, we are an agent in the transaction as we do not physically control the hardware which we sell. Accordingly, starting January 1, 2018, we recognize our hardware revenue net of related cost which reduces both hardware revenue and cost of sales as compared to our accounting prior to 2018. For comparison purposes only, had we implemented ASC 606 using the full retrospective method, we would have also presented hardware revenue net of cost for prior periods as shown below.

	Year Ended December 31,				
				% Change vs. Prior Year	
	2019	**2018**	**2017**	**2019**	**2018**
Hardware Revenue (Pre ASC 606 Adoption)	**$ 44,972**	$ 49,914	43,190	-10%	16%
Cost of hardware	**(32,455)**	(35,947)	(32,205)	-10%	12%
Hardware Revenue, net (Post ASC 606 Adoption)	**$ 12,517**	$ 13,967	$ 10,985	-10%	27%

The majority of hardware sales are derived from our Americas segment. Sales of hardware are largely dependent upon customer-specific desires, which fluctuate.